LECTURE 6

19 September 2023 10:30

1) If $\lim a_n \to l$ (finite & unique) $\to$ cgt

2) If $\lim a_n \to +\infty / -\infty$ div $+\infty$ / div $-\infty$

3) If $\lim a_n \to$ not unique

bdd — osc. finite

unbdd — osc. infinitely

Eg. $\frac{3^n}{n^3}$ $\lim \frac{a_{n+1}}{a_n} : \frac{3^{n+1}}{(n+1)^3} \cdot \frac{n^3}{3^n}$

$= \lim 3 \cdot \frac{n^3}{(n+1)^3} = 3 > 1$

div $+\infty$ $\frac{\frac{3^{n+1}}{(n+1)^3}}{3^n}$

div V $\neq 0$ $\dfrac{\quad}{\dfrac{3^n}{n^2}}$ $\quad$ $(n+1)^3$

Eg. $\left(1+\frac{1}{n}\right)^{n^2}$

lim $(a_n)^{1/n}$: $\left[\left(1+\frac{1}{n}\right)^{n^2}\right]^{1/n} \lim\limits_{=} \left(1+\frac{1}{n}\right)^{n}$

note : $\left[1+f(n)\right]^{g(n)} = e^{f(n)\cdot g(n)}$,

$= e^{1/n \cdot n} = e > 1$

Old friend : $(-1)^n = \{-1, 1, -1, -1 \cdots\}$

subsequences :

$a_{2n} = 1, 1, 1, \ldots$

$a_{2n+1} = -1, -1, -1, \ldots$

$\Big\}$ subseq a_n

$5n, \; 2^n, \; n^2 \; \ldots \; \uparrow$ ~~$\arg\{-n^2\}$~~

a_n seq.	bdd	mono.	cgt	
$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$	$\Rightarrow$
b_n subseq.	bdd	~~non~~mono.	cgt	

<u>Result</u>: If $a_{2n} \to l$ & $a_{2n+1} \to l$ then

$$a_n \to l$$

$(-1)^n n$ $\qquad a_{2n} = n \to \infty$

$a_{2n+1} = -n \to -\infty$

Bolzano Weierstrass Thm: BWT

$\Rightarrow$ Every bdd seq. has a conv. subseq.